RESEARCH BRIEFING · OCTOBER 2026

EUR crypto arbitrage: one year of testing

After every cost, there was no edge to trade.

What remains useful is the data on what EUR crypto execution really costs.

0

trades in a 4-hour live paper run at €100k per leg

−2.3 bps

median BTC price gap, against 17 bps needed to break even

0 of 136

realistic backtest settings that made money

Syed Zee Waqar Hussain · Paper trading only · Public market data · No orders sent

Do price gaps between exchanges survive costs?

1

Watch

BTC and ETH: Bitstamp (EUR) vs Binance (USDT, in EUR). EURC: Bitvavo, Kraken, Bitstamp.

›

2

Price

Measure the gap for the full order size, after crossing both order books.

›

3

Gate

Trade only if the gap beats both exchange fees plus a 1 bp FX cost.

›

4

Fill

Fill both legs 300 ms later. Book losses; count failed fills.

Three kinds of test

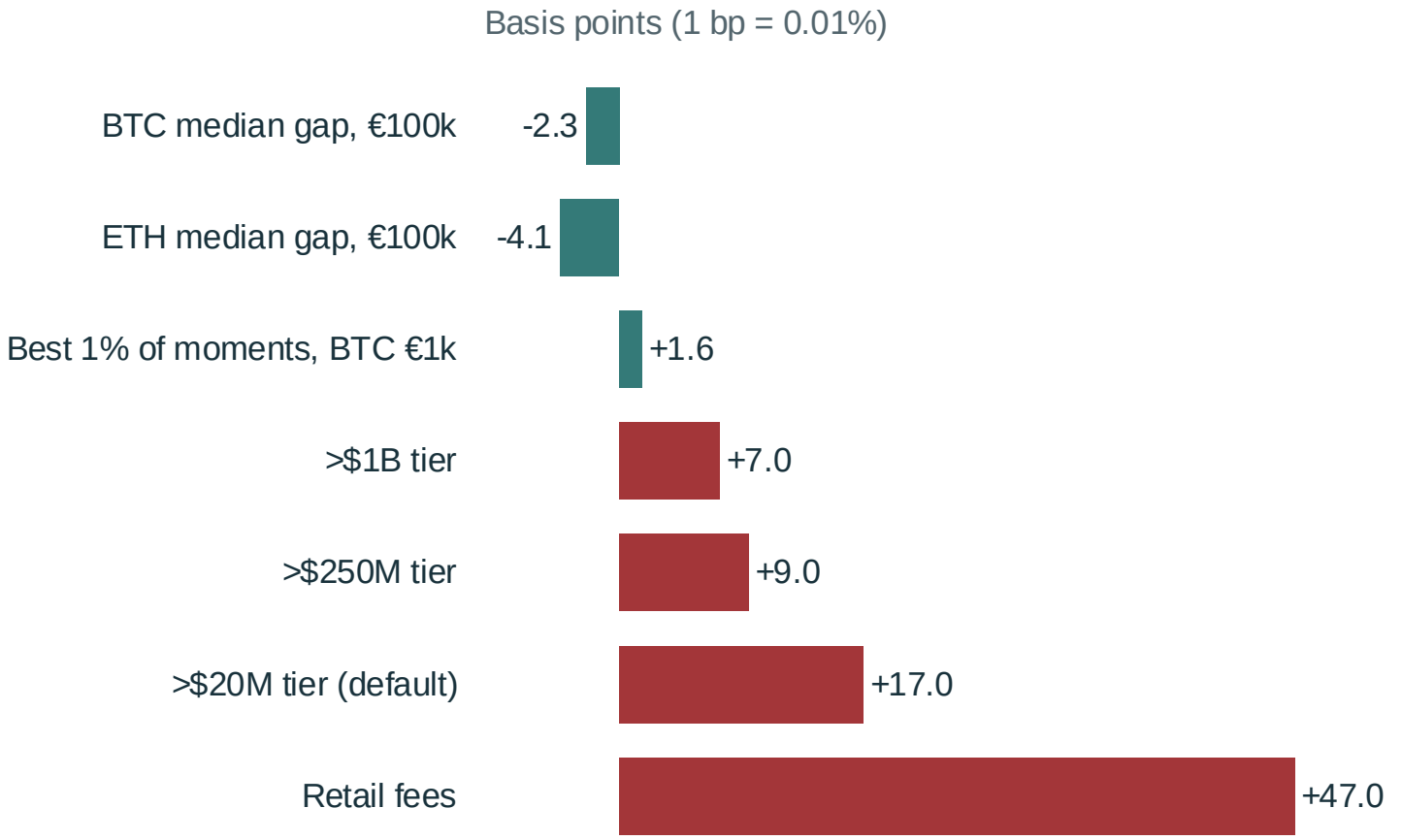
Backtests
One-minute candles, every fee tier, 136 parameter settings

Replays
Recorded order books, fills at 300 / 1,000 / 2,000 ms

Live paper runs
BTC/ETH on 14 Oct (4 h) and EURC on 22 Oct (2.6 h), pass/fail rules set in advance

Sources: bsdex_strategy.py; bsdex_historical_bt.py; scripts/eurc_live_test.py. Bitstamp stands in for BSDEX (its API needs keys); LMAX fees are modelled.

Price gaps are a few bps; costs are 7 to 47 bps



0 trades

in 4 hours of live paper trading at €100k per leg

At €100k the gap was negative 99.6% (BTC) and 99.9% (ETH) of the time.

The 14 moments that beat break-even lasted 3 s at most, and only while one venue’s prices froze.

Live paper run, 14 Oct 2026, 11:04–15:04 UTC, 49,825 snapshots per coin. Gap after crossing both books, before fees. Fee tiers are assumptions.

The profitable backtest was three bugs

BEFORE FIXES

+€276

5 trades · 100% winners · no drawdown

→

AFTER FIXES

−€1,548

7 trades · 0 winners · −€1,740 after funding cost

Bug 1 · Look-ahead

The entry rule read the bar it would be filled on, so it only took trades it already knew would win.

Bug 2 · Wrong FX rate

The daily ECB rate sat 44 bps away from live EUR/USDT, which created gaps that did not exist.

Bug 3 · Instant fills

Paper fills had no delay, so they could never lose. Now: 300 ms and full-depth prices.

On a second week of data, 0 of 136 realistic settings made money.

BTC/EUR, 24 Sep–1 Oct 2026, €100k per leg, >$250M tier (docs/engine-integrity.md). Second week: 7–14 Oct 2026, 1-minute candles, all fee tiers.

Stablecoin test: every pre-set criterion failed

Criterion set in advance	Result	Verdict
Net profit after exit costs above zero	about −€29	FAIL
Market-making P&L at mid of at least zero	−€2.48	FAIL
Median 60 s markout of at least −1 bp	−34 bps	FAIL
Our fills below 20% of venue volume	35%	FAIL
At least 3 sessions, including a weekend	1 session, 2.6 h	NOT MET

The only positive result, a deep bid absorbing one seller sweep, is worth about €100–300 a year.

2 vs 20 bps

Kraken’s spread against its maker fee: never worth quoting

≈6,500 EURC

Bitvavo’s median daily volume over the last 366 days

−39 bps

best cross-venue trade at €10k, after fees

Session 22 Oct 2026, 19:48–22:25 UTC; Bitvavo, Kraken, Bitstamp; €40k per market-making instance. Criteria fixed in docs/eurc_live_test.md before the run.

Not deployed. The measurements are the asset.

Decision

Keep the €40k in the 2.5% deposit.

The market is arbitraged to within a few bps. Paying spreads and fees cannot win; only earning them might.

What would reopen it

- BTC funding above about 7% a year, sustained
- Free, practical par redemption of EURC
- Bitvavo EURC volume above €100k a day

Where the data helps now

1. **Best execution**
   Live spread, depth, freshness and latency per venue (MiCA Art. 78).
2. **Euro stablecoins**
   EURC traded below 0.9925 on 28 of 366 days on Bitvavo.
3. **Model validation**
   A checklist of the errors that made losses look like profit.
4. **Next test worth running**
   Passive quotes with a near-free internal hedge, at real institutional fees.